# Muhammad Arsalan

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**GitHub**: muhammad-Arsalan-247 | **Portfolio**: muhammadarsalan.info |

## Summary

Google-certified Business Intelligence Analyst and AI Engineer with 2+ year of professional experience delivering data-driven solutions across financial services and technology sectors. Developed interactive Power BI dashboards that reduced manual reporting time by 70% and built ETL pipelines improving processing efficiency by 40%. Analyzed 1M+ transactions and deployed machine learning models that improved customer retention by 15%. Proficient in Power BI (DAX, Power Query), PostgreSQL, Python (Pandas, Scikit-learn, TensorFlow, Keras), and LLM integration, with a portfolio of 9 Power BI case studies, 10+ SQL projects, and 54 Python notebooks available on GitHub.

## Work Experience

**Data Analyst / AI Engineer, *PayFast*** (2024 – 2025)

- Analyzed 1M+ financial transactions using PostgreSQL (CTEs, window functions) to identify fraud patterns and revenue opportunities reported directly to leadership.
- Engineered end-to-end ETL pipelines in Python (Pandas, SQL Alchemy) automating data ingestion and transformation, improving pipeline efficiency by 40%.
- Built customer churn prediction model using Scikit-learn Random Forest, achieving a 15% improvement in customer retention through targeted marketing strategies.
- Integrated LLM APIs (Llama) for automated transaction categorization and customer sentiment analysis, reducing manual review workload by 30%.
- Developed anomaly detection system to flag suspicious transactions in real time, reducing false positives by 25% compared to the legacy rule-based system.
- Collaborated with cross-functional teams (product, engineering, finance) to define KPIs and translate business requirements into technical deliverables.

**Power BI Developer / Data Analyst, *Signx Solution*** (2023 – 2024)

- Designed and deployed 3 production Power BI dashboards (HR analytics, sales performance, operational KPIs) serving 50+ stakeholders across departments.
- Reduced manual reporting time by 70% by automating data refresh cycles using Power Query and scheduled pipeline jobs, replacing error-prone Excel workflows.
- Developed advanced DAX measures for cohort analysis, rolling averages, YoY comparisons, and What-If scenario modeling to support executive decision-making.
- Built a star-schema data model integrating 8+ data sources (CRM, ERP, spreadsheets), improving query performance by 35% and ensuring a single source of truth.
- Delivered 9 end-to-end Power BI case studies covering HR attrition, customer churn, healthcare, supply chain, e-commerce, inventory, and mortgage analytics.
- Conducted data literacy workshops for business users, enabling self-service reporting and reducing ad-hoc analytics requests by 50%.

## Skills

**Programming & Tools:** Python (Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, Plotly, TensorFlow, Keras, NLTK, BeautifulSoup, NetworkX), SQL, Power BI (DAX, Power Query), Git, Excel

**Technical:** Data Analysis, Data Visualization, Data Modeling (Star Schema), ETL Pipeline Design, Machine Learning, Deep Learning, NLP & Sentiment Analysis, LLM Integration & Fine-tuning (Llama), Anomaly & Fraud Detection, Computer Vision, Time Series Forecasting, Statistical Analysis (A/B Testing, Hypothesis Testing), Dashboard Creation, Report Writing

## Education

**Bachelor's in Artificial Intelligence**
Karachi, Pakistan

## Certifications

- Google Business Intelligence Professional Certificate — Google / Coursera
- Data Analyst in Power BI — DataCamp
- A Engineer — DataCamp

## Key Projects (Samples at muhammadarsalan.info)

- HR Analytics — Atlas Labs (Power BI, DAX, Star Schema): 4-page HR report analyzing 16.1% attrition rate across departments and tenure segments; implemented conditional columns and custom DAX measures, reducing reporting time by 70%.
- Customer Churn Prediction (Power BI, Python, Scikit-learn): End-to-end churn prediction integrating ML model outputs into a Power BI dashboard; achieved 15% improvement in customer retention through data-driven segmentation.
- Healthcare Analytics Dashboard (Power BI, SQL, DAX): Hospital efficiency dashboard analyzing length-of-stay drivers, resulting in a 12% reduction in average LOS through targeted process improvements.
- Supply Chain Analytics (Power BI, What-If Parameters): Procurement decision-support dashboard with cost-volume curves and scenario modeling, identifying opportunities contributing to 18% cost reduction.
- Inventory & E-Commerce Dashboards (Power BI, Power Query): Retail inventory dashboard with automated reorder alerts reducing excess inventory by 40%; 6-page e-commerce regional sales dashboard with geographic mapping.
- Fraud & Anomaly Detection (Python, Isolation Forest, SQL): Applied IForest model to financial transaction data to identify anomalies indicating potential fraud; reduced false positive rate by 25%.
- LLM Fine-tuning & NLP Pipeline (Python, Llama, Transformers): Fine-tuned Llama model for domain-specific tasks; built sentiment analysis pipeline on automotive review corpus using LLM APIs.
- ETL Pipeline Automation (Python, SQL, Pandas): Built production-grade ETL pipelines for Walmart sales and energy consumption data with automated validation, error handling, and scheduling.
- ML Classification Suite (Python, Scikit-learn): Crop prediction (multi-class), credit card approval, car insurance claim outcomes, and telecom churn prediction models with end-to-end feature engineering and evaluation.
- Reinforcement Learning — Taxi Route Optimization (Python, RL): RL-based taxi route optimization achieving measurable reduction in average travel time across simulated urban grids.
- Over 54 Python, 10+ SQL, and 9 Power BI case study projects including time series forecasting, NLP text mining, Twitter network analysis (NetworkX), computer vision (CNN with Keras), stock portfolio risk modeling, and mortgage analytics — code samples available on GitHub.